

Unit-4 Cyber Crime

4.1 Computer, Cybercrime and Legal Landscape Around the World

Introduction

Computers and the internet have become an important part of daily life. People use digital technologies for banking, shopping, communication, education, healthcare, and business operations. Along with technological growth, cyber crimes are also increasing rapidly.

Cyber criminals misuse computers, networks, and the internet to perform illegal activities such as:

- Hacking
- Data theft
- Online fraud
- Identity theft
- Malware attacks

Different countries create cyber laws and regulations to control cyber crimes and protect digital systems.

1. Meaning of Computer Crime

A computer crime is any illegal activity involving computers, networks, or digital devices.

Examples

- Hacking systems
- Spreading malware
- Stealing passwords
- Online banking fraud

2. Meaning of Cybercrime

Cybercrime refers to crimes committed using:

- Computers
- Mobile phones
- Internet
- Digital networks

Cybercrime can target:

- Individuals
- Businesses
- Governments

3. Types of Cybercrime

A) Hacking

Unauthorized access to computer systems.

B) Phishing

Fake emails or messages used to steal personal information.

C) Identity Theft

Using another person's personal information illegally.

D) Cyber Stalking

Online harassment and threatening behavior.

E) Malware Attacks

Using malicious software to damage systems.

4. Global Growth of Cybercrime

Cybercrime is increasing because of:

- Internet expansion
- Online banking
- Social media usage
- Digital payments
- Cloud computing

5. Legal Landscape Around the World

Different countries have developed cyber laws to fight cybercrime.

5.1 India

Law

Information Technology (IT) Act 2000

Purpose

- Prevent cyber crimes
- Provide legal recognition to electronic records

5.2 United States

Laws

- Computer Fraud and Abuse Act (CFAA)
- Cybersecurity Information Sharing Act

5.3 European Union

Regulation

GDPR (General Data Protection Regulation)

Purpose

Protect user privacy and personal data.

5.4 United Kingdom

Law

Computer Misuse Act

Purpose

Punish unauthorized access and hacking.

6. Challenges in International Cyber Laws

- Cross-border cyber attacks
- Different legal systems
- Difficult identification of attackers
- Lack of global cooperation

7. Importance of Cyber Laws

- Protect users
- Reduce cybercrime

- Ensure data privacy
- Support secure digital systems

8. Conclusion

Cybercrime is a global issue affecting individuals and organizations. Strong cyber laws and international cooperation are important for controlling digital crimes and protecting information systems.

4.2 Criminals Motive of Attackers and Types of Attacks

Introduction

Cyber attackers perform illegal activities for different reasons. Understanding attacker motives and attack types helps organizations improve cyber security and reduce risks.

1. Motives of Cyber Attackers

1.1 Financial Gain

Attackers steal money or financial information.

Examples

- Banking fraud
- Credit card theft
- Ransomware attacks

1.2 Data Theft

Stealing confidential information.

Examples

- Customer databases
- Passwords
- Business secrets

1.3 Revenge

Disgruntled employees attack organizations.

Example

Deleting company files.

1.4 Political Motives

Cyber attacks performed for political purposes.

Example

Attacking government websites

1.5 Cyber Espionage

Stealing sensitive government or business information secretly.

1.6 Fame and Recognition

Some hackers attack systems to gain popularity.

1.7 Terrorism

Using cyber attacks to create fear and disruption.

2. Types of Cyber Attackers

- Script Kiddies
- Professional Hackers
- Insider Threats
- Cyber Terrorists
- State-Sponsored Attackers

3. Types of Cyber Attacks

3.1 Malware Attack

Using malicious software to damage systems.

Types

- Virus
- Worm

- Trojan
- Ransomware

3.2 Phishing Attack

Fake messages used to steal information.

3.3 Password Attack

Cracking passwords to gain access.

3.4 Denial of Service (DoS)

Overloading systems to stop services.

3.5 SQL Injection

Injecting malicious SQL commands into databases.

3.6 Man-in-the-Middle Attack

Intercepting communication secretly.

3.7 Insider Attack

Attack performed by authorized users.

4. Effects of Cyber Attacks

- Financial loss
- Data theft
- Reputation damage
- Service disruption

5. Prevention Methods

- Strong passwords

- Firewalls
- Employee awareness
- Antivirus software
- Multi-factor authentication

6. Conclusion

Cyber attackers have different motives such as money, revenge, politics, and data theft. Understanding attack methods helps organizations improve security and reduce cyber risks.

4.3 Cyber Threats – Cyber Warfare

Introduction

Cyber warfare refers to the use of cyber attacks by nations or organized groups to damage another country's computer systems, networks, or critical infrastructure.

Cyber warfare is considered a modern form of warfare using digital technology instead of traditional weapons.

1. Meaning of Cyber Warfare

Cyber warfare is the use of cyber attacks to:

- Disrupt operations
- Damage infrastructure
- Steal information
- Create fear

2. Targets of Cyber Warfare

- Government systems
- Military networks
- Power grids
- Banking systems
- Communication networks

3. Types of Cyber Warfare Attacks

3.1 Malware Attacks

Using malicious software against national systems.

3.2 DDoS Attacks

Flooding systems with traffic.

3.3 Espionage Attacks

Stealing secret information.

3.4 Infrastructure Attacks

Attacking:

- Electricity systems
- Water supply systems
- Transportation systems

4. Features of Cyber Warfare

- Difficult to identify attackers
- Fast execution
- Global impact
- Low physical risk

5. Real-Life Examples

A) Stuxnet Attack

Malware targeting Iran's nuclear systems.

B) Attacks on Government Websites

Countries targeting each other's digital infrastructure.

6. Impact of Cyber Warfare

- National security threats
- Economic damage
- Public fear
- Communication disruption

7. Prevention and Defense

- Strong cyber defense systems
- Security monitoring
- Encryption
- International cooperation

8. Conclusion

Cyber warfare is a serious global threat. Nations must strengthen cyber security systems and develop effective cyber defense strategies to protect critical infrastructure.

4.4 Comprehensive Cyber Security Policy

Introduction

A cyber security policy is a set of rules and guidelines that help organizations protect digital systems, networks, and information from cyber threats.

It defines:

- Security responsibilities
- Access control rules
- Data protection methods
- Incident response procedures

1. Objectives of Cyber Security Policy

- Protect confidential data
- Reduce cyber risks
- Ensure legal compliance
- Maintain business continuity

2. Components of Cyber Security Policy

2.1 Access Control Policy

Defines who can access systems.

2.2 Password Policy

Rules for creating strong passwords.

2.3 Data Protection Policy

Protects sensitive information.

2.4 Network Security Policy

Protects organizational networks.

2.5 Incident Response Policy

Steps to handle cyber incidents.

2.6 Backup Policy

Rules for data backup and recovery.

3. Importance of Cyber Security Policy

- Improves security awareness
- Reduces cyber attacks
- Defines responsibilities
- Protects business operations

4. Best Practices

- Regular updates
- Employee training
- Strong authentication
- Security monitoring

5. Conclusion

A comprehensive cyber security policy helps organizations protect systems, data, and networks from cyber threats and ensures secure business operations.

4.5 Cybercrimes Targeting Computer Systems and Mobiles

Introduction

Computers and mobile devices are major targets of cyber criminals because they store valuable personal and financial information.

1. Cybercrimes Targeting Computers

1.1 Hacking

Unauthorized access to systems.

1.2 Malware Infection

Viruses and ransomware damaging systems.

1.3 Data Theft

Stealing confidential information.

1.4 Software Piracy

Illegal copying of software.

2. Cybercrimes Targeting Mobiles

2.1 Mobile Malware

Malicious mobile applications.

2.2 SIM Swapping

Attackers gain control of mobile numbers.

2.3 Fake Apps

Apps designed to steal information.

2.4 Mobile Banking Fraud

Stealing banking details through mobile attacks.

3. Prevention Methods

- Install antivirus
- Download apps from trusted sources
- Use strong passwords
- Update software regularly

4. Conclusion

Computers and mobile devices are common targets for cyber criminals. Proper security measures are necessary to protect devices and data.

4.6 Online Scams and Frauds

Introduction

Online scams and frauds are illegal activities performed using the internet to cheat people and steal money or personal information.

1. Common Types of Online Scams

1.1 Phishing Scam

Fake emails or messages requesting personal details.

1.2 Lottery Scam

Fake messages claiming prize money.

1.3 Online Shopping Fraud

Fake websites selling products.

1.4 Job Scam

Fake job offers demanding money

1.5 Banking Fraud

Stealing bank account information.

2. Effects of Online Frauds

- Financial loss
- Identity theft
- Emotional stress

3. Prevention Methods

- Verify websites
- Avoid sharing OTPs
- Use secure payment methods

- Be careful with unknown links

4. Conclusion

Online scams are increasing rapidly. User awareness and cyber security practices help reduce fraud risks.

4.7 Cybercrime and Punishments

Introduction

Governments create cyber laws to punish cyber criminals and protect digital systems.

1. Common Cybercrimes

- Hacking
- Identity theft
- Data theft
- Cyber stalking
- Phishing

2. Punishments Under Cyber Laws

Punishments may include:

- Imprisonment
- Monetary fines
- Both

3. Cybercrime Punishments in India

Under the IT Act 2000:

A) Hacking

Punishable with imprisonment and fine.

B) Identity Theft

Illegal use of another person's information.

C) Cyber Terrorism

Severe punishment including long imprisonment.

4. Importance of Punishments

- Discourages cyber crimes
- Protects users
- Maintains law and order

5. Conclusion

Strict cyber laws and punishments are important for controlling cyber crimes and protecting digital systems.

4.8 Cyber Laws and Legal and Ethical Aspects Related to New Technologies – AI/ML, IoT, Blockchain, Darknet and Social Media

Introduction

Modern technologies such as AI, Machine Learning, IoT, Blockchain, Darknet, and Social Media provide many benefits but also create legal and ethical challenges.

Cyber laws and ethical practices help ensure safe and responsible use of technology.

1. AI and Machine Learning

Legal Issues

- Data privacy violations
- Biased decision-making

Ethical Issues

- Lack of transparency
- Job replacement concerns

2. Internet of Things (IoT)

Legal Issues

- Weak device security
- Data collection risks

Ethical Issues

- Privacy invasion
- Unauthorized monitoring

3. Blockchain Technology

Legal Issues

- Cryptocurrency regulations
- Illegal transactions

Ethical Issues

- Misuse for criminal activities

4. Darknet

Meaning

Hidden part of the internet not indexed by normal search engines.

Risks

- Illegal trade
- Drug trafficking
- Cybercrime activities

5. Social Media

Legal Issues

- Fake news
- Cyber bullying
- Privacy violations

Ethical Issues

- Misinformation
- Online harassment

6. Importance of Ethical Technology Use

- Protect user rights
- Maintain privacy

- Prevent misuse
- Promote trust

7. Conclusion

New technologies create both opportunities and risks. Cyber laws and ethical guidelines are necessary for responsible and secure technology usage.

4.9 Case Studies on Online Scams and Frauds/Cybercrime and Punishments

Introduction

Case studies help understand real cyber crimes, fraud methods, and legal punishments.

Case Study 1: Online Banking Fraud

Situation

A user receives a fake bank SMS asking for OTP verification.

Attack

The victim shares OTP details.

Result

Money stolen from bank account.

Punishment

Cyber police investigate under IT Act provisions.

Lesson

Never share OTPs or passwords.

Case Study 2: Ransomware Attack

Situation

A hospital's computer systems become locked.

Attack

Ransomware encrypts patient records.

Result

Hospital operations disrupted.

Punishment

Attackers face severe cybercrime charges.

Lesson

Regular backups and updates are important.

Case Study 3: Social Media Fake Profile

Situation

A fake profile is created using another person's photos.

Attack

Identity misuse and online harassment.

Result

Victim faces emotional stress.

Punishment

Legal action under cyber laws.

Lesson

Use privacy settings and report fake accounts.

Case Study 4: Online Shopping Fraud

Situation

Customer orders products from fake website.

Attack

Payment taken but products not delivered.

Result

Financial loss.

Lesson

Shop only from trusted websites.

Conclusion

Case studies show how cyber crimes affect individuals and organizations. Awareness, strong security practices, and proper cyber laws help reduce cybercrime and protect users.

